CSE-476: Introduction to Natural Language Processing
Quiz 4

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

Q1. What is one of the key limitations of RNN?

(A) They cannot process sequences
(B) They are too computationally cheap
(C) They cannot use embeddings
(D) They struggle with long-range dependencies

Q2. Which model type is generally best for text classification tasks (e.g., sentiment analysis)?

(A) Encoder-only
(B) Decoder-only
(C) Encoder-decoder
(D) None of the above

Q3. What does self-attention allow a model to do?

(A) Process only one token at a time
(B) Attend to other tokens within the same sequence
(C) Ignore context entirely
(D) Only use previous tokens

Q4. What is a key characteristic of transformers compared to RNNs?

(A) They process tokens strictly sequentially
(B) They rely on recurrence
(C) They use self-attention instead of recurrence
(D) They cannot handle sequences

Q5. Which model would work best for a task like summarization?

(A) Encoder-only
(B) Decoder-only

(C) Encoder-decoder
(D) None

Q6. What is a key advantage of Rotary Positional Embeddings?

(A) They require no training data
(B) They eliminate attention
(C) They only work for short sequences
(D) They preserve relative positions and generalize well

Q7. What type of transformer is BERT?

(A) Encoder-only
(B) Decoder-only
(C) Encoder-decoder
(D) Recurrent transformers

Q8. What is the main objective used to pre-train BERT?

(A) Next-word prediction
(B) Image classification
(C) Masked language modeling
(D) Reinforcement learning

Q9. Which type of positional encoding uses sine and cosine functions?

(A) Learned positional encoding
(B) Relative positional encoding
(C) Rotary positional encoding
(D) Sinusoidal positional encoding

Q10. What is multi-head attention used for?

(A) Reducing model parameters
(B) Capturing diverse relationships in data
(C) Eliminating positional encoding
(D) Replacing embeddings

End of Paper